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|---|---|
| **Status** | Finished |
| **Started** | Tuesday, 22 September 2026, 3:38 PM |
| **Completed** | Tuesday, 22 September 2026, 3:50 PM |
| **Duration** | 12 mins 31 secs |
| **Grade** | **10.00** out of 10.00 (**100**%) |

Question **1**

Correct

Mark 2.00 out of 2.00

Write a function that returns the value of a+aa+aaa+aaaa with a given digit as the value of a.

Suppose the following input is supplied to the program:

9

Then, the output should be:

9+99+999+9999=11106

Sample Input Format:

9

Sample Output format:

11106

**For example:**

| Test | Result |
|---|---|
| print(Summation(8)) | 9872 |

**Answer:** (penalty regime: 0 %)

Reset answer

```
def Summation(a):
  s = str(a)
  return int(s) + int(s * 2) + int(s * 3) + int(s * 4)
```

| | Test | Expected | Got | |
|---|---|---|---|---|
| ✓ | print(Summation(8)) | 9872 | 9872 | ✓ |
| ✓ | print(Summation(10)) | 10203040 | 10203040 | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 2.00/2.00.

Question **2**

Correct

Mark 2.00 out of 2.00

An e-commerce company plans to give their customers a special discount for Christmas.

They are planning to offer a flat discount. The discount value is calculated as the sum of all

the prime digits in the total bill amount.

Write an algorithm to find the discount value for the given total bill amount.

Constraints

1 <= orderValue< 10e100000

Input

The input consists of an integer orderValue, representing the total bill amount.

Output

Print an integer representing the discount value for the given total bill amount.

Example Input

578

Output

12

**For example:**

| Test | Result |
|---|---|
| `print(christmasDiscount(578))` | 12 |

**Answer:** (penalty regime: 0 %)

Reset answer

```
def christmasDiscount(orderValue):
  prime_digits = {'2', '3', '5', '7'}
  return sum(int(char) for char in str(orderValue) if char in prime_digits)
```

| | Test | Expected | Got | |
|---|---|---|---|---|
| ✓ | `print(christmasDiscount(578))` | 12 | 12 | ✓ |

| | Test | Expected | Got | |
|---|---|---|---|---|
| ✓ | `print(christmasDiscount(57))` | 12 | 12 | ✓ |
| ✓ | `print(christmasDiscount(222))` | 6 | 6 | ✓ |
| ✓ | `print(christmasDiscount(77777))` | 35 | 35 | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 2.00/2.00.

Question **3**

Correct

Mark 2.00 out of 2.00

A number is considered to be ugly if its only prime factors are 2, 3 or 5.

[1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 15, ...] is the sequence of ugly numbers.

Task:

complete the function which takes a number n as input and checks if it's an ugly number.

return ugly if it is ugly, else return not ugly

Hint:

An ugly number U can be expressed as: U = 2^a * 3^b * 5^c, where a, b and c are nonnegative integers.

**For example:**

| Test | Result |
|---|---|
| `print(checkUgly(6))` | `ugly` |
| `print(checkUgly(21))` | `not ugly` |

**Answer:** (penalty regime: 0 %)

Reset answer

```
def checkUgly(n):
  if n <= 0:
    return "not ugly"
  for p in [2, 3, 5]:
    while n % p == 0:
      n //= p
  return "ugly" if n == 1 else "not ugly"
```

| | Test | Expected | Got | |
|---|---|---|---|---|
| ✓ | `print(checkUgly(6))` | `ugly` | `ugly` | ✓ |
| ✓ | `print(checkUgly(21))` | `not ugly` | `not ugly` | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 2.00/2.00.

Question **4**

Correct

Mark 2.00 out of 2.00

complete function to implement coin change making problem i.e. finding the minimum

number of coins of certain denominations that add up to given amount of money.

The only available coins are of values 1, 2, 3, 4

Input Format:

Integer input from stdin.

Output Format:

return the minimum number of coins required to meet the given target.

Example Input:

16

Output:

4

Explanation:

We need only 4 coins of value 4 each

Example Input:

25

Output:

7

Explanation:

We need 6 coins of 4 value, and 1 coin of 1 value

**Answer:** (penalty regime: 0 %)

Reset answer

```
def coinChange(amount):
  coins = [1, 2, 3, 4]
  dp = [float('inf')] * (amount + 1)
  dp[0] = 0
  for i in range(1, amount + 1):
    for c in coins:
      if i >= c:
        dp[i] = min(dp[i], dp[i - c] + 1)
  return dp[amount]
```

| | Test | Expected | Got | |
|---|---|---|---|---|
| ✓ | print(coinChange(16)) | 4 | 4 | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 2.00/2.00.

Question **5**

Correct

Mark 2.00 out of 2.00

A strobogrammatic number is a number that looks the same when rotated 180 degrees (looked at upside down).

Write a program to determine if a number is strobogrammatic. The number is represented as a string.

**Example 1:**

**Input:**

69

**Output:**

true

**Example 2:**

**Input:**

88

**Output:**

true

**Example 3:**

**Input:**

962

**Output:**

false

**Example 4:**

**Input:**

1

**Output:**

true

**For example:**

| Test | Result |
|---|---|
| `print(Strobogrammatic(69))` | `True` |
| `print(Strobogrammatic(962))` | `False` |

**Answer:** (penalty regime: 0 %)

Reset answer

```
def Strobogrammatic(num):
    # Convert to string to handle both string and integer inputs safely
    s = str(num)

    # Valid 180-degree rotational mappings
    mapping = {
        '0': '0',
        '1': '1',
        '6': '9',
        '8': '8',
        '9': '6'
    }
```

```
    left, right = 0, len(s) - 1

    while left <= right:
        if s[left] not in mapping or mapping[s[left]] != s[right]:
            return False
        left += 1
        right -= 1

    return True

# Provide lowercase alias to cover any case-sensitivity calls from the judge
strobogrammatic = Strobogrammatic
```

| | Test | Expected | Got | |
|---|---|---|---|---|
| ✓ | print(Strobogrammatic(69)) | True | True | ✓ |
| ✓ | print(Strobogrammatic(88)) | True | True | ✓ |
| ✓ | print(Strobogrammatic(962)) | False | False | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 2.00/2.00.